I.20 · Turn two sides into one longer straight length

In any triangle, any two sides taken together are greater than the remaining side.

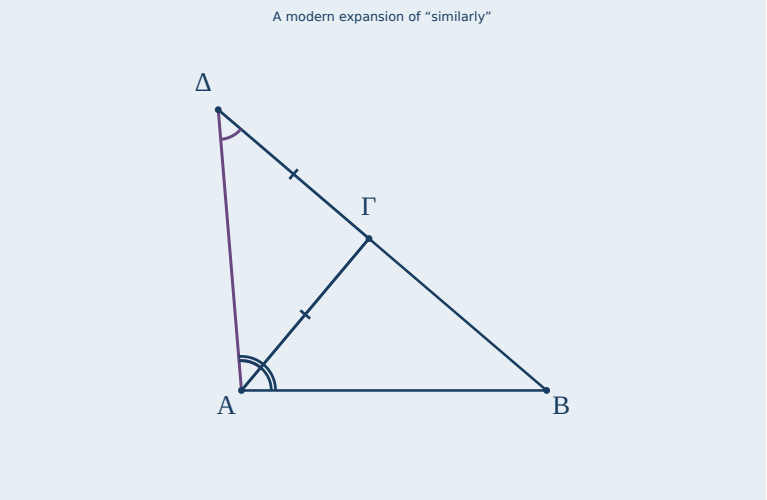


Plate at step 8: Construct the other two comparisons.

A modern expansion of “similarly”

Continue BΓ through Γ to Δ; copy ΓA to ΓΔ and join ΔA. The angle at A in ΔAB contains the isosceles angle as a proper part. I.19 makes ΔB longer than BA. The original passage remains unchanged.

`ΓΔ = ΓA (copied length) · ∠ΓΔA = ∠ΓAΔ (I.5) · ∠BAΔ > ∠BΔA (whole > part) · ΔB > BA (I.19) · ΔB = BΓ + ΓA > BA`

The proposition

Παντὸς τριγώνου αἱ δύο πλευραὶ τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι.	In quouis triangulo duo latera reliquo maiora sunt quoquo modo coniuncta.	In any triangle, any two sides taken together are greater than the remaining side.

1. Name all three comparisons Given

Ἔστω γὰρ τρίγωνον τὸ ΑΒΓ· λέγω, ὅτι τοῦ ΑΒΓ τριγώνου αἱ δύο πλευραὶ τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι, αἱ μὲν ΒΑ, ΑΓ τῆς ΒΓ, αἱ δὲ ΑΒ, ΒΓ τῆς ΑΓ, αἱ δὲ ΒΓ, ΓΑ τῆς ΑΒ.	Sit enim triangulus ΑΒΓ. dico, in triangulo ΑΒΓ duo latera reliquo maiora esse quoquo modo coniuncta, ΒΑ + ΑΓ > ΒΓ, ΑΒ + ΒΓ > ΑΓ, ΒΓ + ΓΑ > ΑΒ.	Let ΑΒΓ be a triangle. I say that any two sides together exceed the remaining side: ΒΑ and ΑΓ exceed ΒΓ; ΑΒ and ΒΓ exceed ΑΓ; ΒΓ and ΓΑ exceed ΑΒ.

μὲν … δὲ … δέ distributes the claim over three pairs. Latin displays the three inequalities directly. These are the goals, not results proved at the opening step.

Reason: The task and its given objects A given object is part of the problem. It has not been constructed by this proof. A finite straight line here means a line segment.

Printed source: Greek pp. 48; Latin pp. 49.

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2. Continue BA beyond A Construct

Διήχθω γὰρ ἡ ΒΑ ἐπὶ τὸ Δ σημεῖον,	educatur enim ΒΑ ad Δ punctum,	For let ΒΑ be continued to the point Δ,

The order ΒΑ matters: continue from Β through Α. The name Δ anticipates the point whose distance from Α the next command fixes. This plate first shows the continuation as a ray.

Reason: Postulate 2 · continue the side Continue the specified side along its own straight line. This supplies an exterior angle; it does not assume the later theorem about the sum of a triangle’s angles.

Printed source: Greek pp. 48; Latin pp. 49.

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3. Copy ΓΑ to ΑΔ, then join ΔΓ Construct

καὶ κείσθω τῇ ΓΑ ἴση ἡ ΑΔ, καὶ ἐπεζεύχθω ἡ ΔΓ.	et ponatur ΑΔ = ΓΑ, et ducatur ΔΓ.	and let ΑΔ be laid off equal to ΓΑ, and let ΔΓ be joined.

κείσθω and ἐπεζεύχθω are separate commands, as are ponatur and ducatur. One establishes an equal length; the other joins the newly fixed point to Γ.

Reason: Copy ΓΑ onto the continuation at A, then join ΔΓ Euclid asks for ΑΔ = ΓΑ on the continuation of ΒΑ. Here I.2 supplies a length at A, and a circle at A cuts that ray at Δ. Join ΔΓ by postulate 1. The auxiliary I.2 work is collapsed; the earlier lesson exposes it.

Printed source: Greek pp. 48; Latin pp. 49.

4. Use the small isosceles triangle Prove

Ἐπεὶ οὖν ἴση ἐστὶν ἡ ΔΑ τῇ ΑΓ, ἴση ἐστὶ καὶ γωνία ἡ ὑπὸ ΑΔΓ τῇ ὑπὸ ΑΓΔ·	iam quoniam ΔA = AΓ, erit etiam ∠ AΔΓ = AΓΔ.	Since ΔA equals AΓ, angle AΔΓ also equals AΓΔ.

The repeated ἴση links a side equality to an angle equality through I.5. ΔA and AΔ name the same segment; the reversed endpoint order does not change its length.

Reason: I.5 · equal sides give equal base angles Identify the two equal sides of the small triangle. Their opposite angles are equal by I.5; these are the angles subsequently compared with a larger angle.

Printed source: Greek pp. 48; Latin pp. 49.

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5. Compare the whole angle with its part Prove

μείζων ἄρα ἡ ὑπὸ ΒΓΔ τῆς ὑπὸ ΑΔΓ.	itaque ∠ ΒΓΔ > AΔΓ.	Therefore ΒΓΔ is greater than AΔΓ.

The short sentence compresses a substitution: ΒΓΔ contains AΓΔ, and AΓΔ equals AΔΓ. Follow both angle names before accepting ἄρα / itaque.

Reason: Common notion 5 · ΒΓΔ contains AΓΔ Ray ΓA lies strictly inside angle ΒΓΔ. Thus ΒΓΔ exceeds AΓΔ, which equals AΔΓ by I.5. Since B, A and Δ are in that order on a line, rays ΔA and ΔB coincide: AΔΓ is also ΒΔΓ.

Printed source: Greek pp. 48; Latin pp. 49.

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6. Use I.19 in the larger triangle Prove

καὶ ἐπεὶ τρίγωνόν ἐστι τὸ ΔΓΒ μείζονα ἔχον τὴν ὑπὸ ΒΓΔ γωνίαν τῆς ὑπὸ ΒΔΓ, ὑπὸ δὲ τὴν μείζονα γωνίαν ἡ μείζων πλευρὰ ὑποτείνει, ἡ ΔΒ ἄρα τῆς ΒΓ ἐστι μείζων.	et quoniam triangulus est ΔΓΒ maiorem habens angulum ΒΓΔ angulo ΒΔΓ, sub maiore autem angulo maius latus subtendit, erit ΔΒ > ΒΓ.	Since triangle ΔΓΒ has angle ΒΓΔ greater than ΒΔΓ, and the side opposite the greater angle is greater, ΔΒ is greater than ΒΓ.

ὑπὸ δὲ τὴν μείζονα γωνίαν … repeats I.19’s conclusion inside the new proof. Track the change from triangle ΔAΓ to triangle ΔΓΒ.

Reason: I.19 · use the angle order in triangle ΔΓΒ In ΔΓΒ, angle ΒΓΔ exceeds ΒΔΓ. The side opposite the greater angle, ΔΒ, therefore exceeds ΒΓ. This is why I.19 was needed before the triangle inequality.

Printed source: Greek pp. 48; Latin pp. 49.

7. Replace ΔA with its equal AΓ Prove

ἴση δὲ ἡ ΔΑ τῇ ΑΓ· μείζονες ἄρα αἱ ΒΑ, ΑΓ τῆς ΒΓ·	uerum ΔA = AΓ. itaque BA + AΓ > BΓ.	But ΔA equals AΓ. Therefore BA and AΓ together are greater than BΓ.

The plural αἱ ΒΑ, ΑΓ names the two sides taken together. Latin writes the addition explicitly. The whole ΔΒ contains BA plus AΔ because A lies between its endpoints.

Reason: Replace the copied segment in the whole length A lies between B and Δ, so ΔB = BA + AΔ. Replace AΔ by its constructed equal AΓ. Since ΔB > BΓ, we obtain BA + AΓ > BΓ. Heiberg spells out the segment addition in a Latin footnote; it is explained here rather than inserted into Euclid's text.

Printed source: Greek pp. 48; Latin pp. 49.

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8. Construct the other two comparisons Prove

ὁμοίως δὴ δείξομεν, ὅτι καὶ αἱ μὲν ΑΒ, ΒΓ τῆς ΓΑ μείζονές εἰσιν, αἱ δὲ ΒΓ, ΓΑ τῆς ΑΒ.	similiter demonstrabimus, esse etiam AB + BΓ > ΓA et BΓ + ΓA > AB.	Similarly we shall show that AB and BΓ together exceed ΓA, and that BΓ and ΓA together exceed AB.

ὁμοίως / similiter requests the same reasoning with different roles. The selector redraws the copied segment and the containing angle for each pair, while leaving the source sentence unchanged.

Reason: Repeat the argument with the other pair The source says similarly. The selector expands the two remaining cases by reassigning the roles of A, B and Γ. Δ is always the new extension point. These extra drawings are modern teaching expansions, not additional source diagrams.

Printed source: Greek pp. 48; Latin pp. 49.

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9. Conclude the triangle inequality Conclude

Παντὸς ἄρα τριγώνου αἱ δύο πλευραὶ τῆς λοιπῆς μείζονές εἰσι πάντῃ μεταλαμβανόμεναι· ὅπερ ἔδει δεῖξαι.	Ergo in quouis triangulo duo latera reliquo maiora sunt quoquo modo coniuncta; quod erat demonstrandum.	Therefore, in any triangle, any two sides taken together exceed the remaining side. This is what had to be shown.

τῆς λοιπῆς leaves πλευρᾶς understood: 'than the remaining side'. Latin reliquo likewise leaves latere understood. πάντῃ / quoquo modo covers all three choices.

Reason: Repeat the argument with the other pair The source says similarly. The selector expands the two remaining cases by reassigning the roles of A, B and Γ. Δ is always the new extension point. These extra drawings are modern teaching expansions, not additional source diagrams.

Printed source: Greek pp. 50; Latin pp. 51.

Prove the shorter route instead of assuming it

A two-side route becomes the straight segment BΔ by copying AΓ beyond A. The comparison ΔB > BΓ follows from angles in the larger triangle, using I.5 and I.19. The other two inequalities require the same construction at the other vertices; the selector draws them explicitly.

Definitions and starting permissions used in this proof

Postulate 2 · β′ / II · Extend a straight line

Καὶ πεπερασμένην εὐθεῖαν κατὰ τὸ συνεχὲς ἐπ’ εὐθείας ἐκβαλεῖν.	Et ut recta linea terminata in directum educatur in continuum.	And that a finite straight line can be extended continuously in a straight line.

πεπερασμένην / terminata identifies a bounded segment. ἐπ’ εὐθείας / in directum keeps the extension straight: the command does not permit a turn at the endpoint.

Printed source: Greek pp. 8; Latin pp. 9.

Postulate 1 · α′ / I · Join two points

Ἠιτήσθω ἀπὸ παντὸς σημείου ἐπὶ πᾶν σημεῖον εὐθεῖαν γραμμὴν ἀγαγεῖν.	Postuletur, ut a quouis puncto ad quoduis punctum recta linea ducatur.	Let it be postulated that a straight line can be drawn from any point to any point.

Ἠιτήσθω / Postuletur means ‘let it be requested / postulated’. The Greek then uses an infinitive, ἀγαγεῖν (‘to draw’); Latin uses ut … ducatur, ‘that … be drawn’. The following postulates continue the opening request.

Printed source: Greek pp. 8; Latin pp. 9.

Postulate 3 · γ′ / III · Draw a circle

Καὶ παντὶ κέντρῳ καὶ διαστήματι κύκλον γράφεσθαι.	Et ut quouis centro radioque circulus describatur.	And that a circle can be drawn with any centre and distance.

κέντρῳ and διαστήματι are datives; Latin centro and radio are ablatives. The two languages attach the circle-drawing operation to a centre and a distance. Compare the fuller command Κέντρῳ … διαστήματι … γεγράφθω in I.1.

Printed source: Greek pp. 8; Latin pp. 9.

Common notion 5 in the usual list · η′ / VIII in Heiberg · Whole and proper part

Καὶ τὸ ὅλον τοῦ μέρους μεῖζόν [ἐστιν].	Et totum parte maius est.	And the whole [is] greater than the part.

Greek uses the genitive τοῦ μέρους after the comparative μεῖζον; Latin uses the ablative parte after maius. Both mean ‘greater than the part’. The brackets around [ἐστιν] belong to the printed Greek.

Printed source: Greek pp. 10; Latin pp. 11.

Greek and Latin: Euclidis opera omnia, vol. I, ed. and Latin translation J. L. Heiberg, Leipzig: Teubner, 1883, printed pp. 2–51. Public domain. University of Toronto scan. English, reading notes and redrawn

coordinate diagrams: Stoicheia (2026), MIT. Greek ϑ is normalized to θ; line breaks and inline references are omitted. Latin u/v spelling is retained. Heiberg's square brackets in I.1, I.4, I.8, I.13 and I.15 are retained. Printed angle and triangle symbols are represented as ∠ and △. The English is a reading translation, including explicit explanations of implied intersections.

The diagram is numerical evidence, not a formal proof. Length units and orientation are modern teaching choices.